Spatial Prisoner's Dilemma

> A computational exploration of Nowak and May's spatial Prisoner's Dilemma, showing how local interaction on a lattice allows cooperation to survive through cluster formation — even without memory, reciprocity, or reputation.

Learning objectives

- Explain why spatial structure can sustain cooperation in the Prisoner's Dilemma without reciprocity or punishment.
- Implement a lattice-based spatial PD with Moore neighborhood interaction and imitation dynamics in R.
- Visualize the emergence and persistence of cooperator clusters on a grid.
- Analyze how the fraction of cooperators evolves over time under different payoff parameters.

Motivation

In @ref(sec-axelrod-tournament), cooperation emerged because players could remember past interactions and condition their behavior on history. But many biological and social systems lack this luxury. Bacteria, plants, and even commuters on a highway interact with nearby neighbors without tracking individual histories. Can cooperation survive when there is no memory, no reputation, and no punishment — only spatial structure?

In 1992, Martin Nowak and Robert May [-@nowak1992] showed that the answer is yes. They placed players on a two-dimensional grid where each cell plays a one-shot Prisoner's Dilemma with its immediate neighbors. After each round, every cell adopts the strategy of whichever neighbor (including itself) scored highest. The result was striking: cooperators formed compact clusters that could resist invasion by defectors, producing intricate, dynamic spatial patterns.

This chapter replicates their model in R. We will build a lattice simulation, watch cooperation clusters emerge, and measure how population-level cooperation rates evolve over time.

Theory

From well-mixed to spatial populations

Standard evolutionary game theory assumes a **well-mixed population**: every individual is equally likely to interact with every other. Under this assumption, defection dominates in a one-shot Prisoner's Dilemma because a defector always earns more than a cooperator against any given opponent.

Spatial structure breaks this assumption. When interaction is **local** — each player meets only its geographic neighbors — cooperators can form clusters. Within a cluster, cooperators interact mainly with other cooperators, earning the mutual cooperation payoff R many times. A defector on the boundary of such a cluster exploits its cooperator neighbors but has fewer cooperator neighbors than cooperators in the cluster interior do.

The Nowak–May model

The model works as follows:

1. **Grid**: An $N \times N$ lattice with periodic boundary conditions (a torus, so edges wrap around).
2. **Strategies**: Each cell is either a cooperator (C) or a defector (D).

3. **Interaction**: Each cell plays a one-shot PD with each of its eight Moore neighbors and itself (nine games total). The cell's score is the sum of payoffs from all nine games.
4. **Update**: Synchronously, every cell adopts the strategy of the highest-scoring cell in its Moore neighborhood (including itself). Ties are broken in favor of the current occupant.

With canonical PD payoffs ($T > R > P > S$), the dynamics depend critically on the temptation-to-defect parameter $b = T/R$. For moderate values of b, cooperators and defectors coexist in dynamic spatial patterns. For very high b, defection takes over; for low b, cooperation dominates.

Why clusters protect cooperators

Consider a 3×3 block of cooperators surrounded by defectors. The cooperator at the center plays nine cooperators (including itself), scoring $9R$. A defector adjacent to this block has at most three cooperator neighbors, scoring at most $3T + 6P$. When $9R > 3T + 6P$ — that is, when $b < 3(R - P/R) + P$ approximately — the interior cooperator outscores the boundary defector, and the cluster holds. This is the geometric logic behind spatial cooperation.

Implementation in R

Grid initialization and neighbor scoring

```r
initialize_grid <- function(n, prob_coop = 0.5) {
  matrix(sample(c("C", "D"), n * n, replace = TRUE,
                prob = c(prob_coop, 1 - prob_coop)),
         nrow = n, ncol = n)
}

get_moore_neighbors <- function(grid, i, j) {
  n <- nrow(grid)
  rows <- ((i - 2):(i)) %% n + 1
  cols <- ((j - 2):(j)) %% n + 1
  grid[rows, cols]
}

score_cell <- function(grid, i, j) {
  neighbors <- get_moore_neighbors(grid, i, j)
  focal <- grid[i, j]
  total <- 0
  for (s in as.vector(neighbors)) {
    total <- total + play_round(focal, s)[1]
  }
  total
}

compute_scores <- function(grid) {
  n <- nrow(grid)
  scores <- matrix(0, nrow = n, ncol = n)
  for (i in seq_len(n)) {
    for (j in seq_len(n)) {
      scores[i, j] <- score_cell(grid, i, j)
    }
```

```
  }
  scores
}
```

Update rule: imitate the best neighbor

```
update_grid <- function(grid, scores) {
  n <- nrow(grid)
  new_grid <- grid
  for (i in seq_len(n)) {
    for (j in seq_len(n)) {
      rows <- ((i - 2):(i)) %% n + 1
      cols <- ((j - 2):(j)) %% n + 1
      neighborhood_scores <- scores[rows, cols]
      best <- which(neighborhood_scores == max(neighborhood_scores),
                    arr.ind = TRUE)
      if (nrow(best) == 1) {
        bi <- rows[best[1, 1]]
        bj <- cols[best[1, 2]]
        new_grid[i, j] <- grid[bi, bj]
      }
      # Tie: keep current strategy (new_grid[i,j] already equals grid[i,j])
    }
  }
  new_grid
}
```

Running the simulation

```
run_spatial_pd <- function(n, generations, prob_coop = 0.5, seed = 42) {
  set.seed(seed)
  grid <- initialize_grid(n, prob_coop)

  history <- vector("list", generations + 1)
  history[[1]] <- grid
  coop_frac <- numeric(generations + 1)
  coop_frac[1] <- mean(grid == "C")

  for (g in seq_len(generations)) {
    scores <- compute_scores(grid)
    grid <- update_grid(grid, scores)
    history[[g + 1]] <- grid
    coop_frac[g + 1] <- mean(grid == "C")
  }

  list(history = history, coop_frac = coop_frac)
}
```

Visualizing a grid snapshot

```
grid_to_df <- function(grid, gen) {
  n <- nrow(grid)
  expand.grid(row = seq_len(n), col = seq_len(n)) |>
    mutate(strategy = as.vector(grid),
           generation = gen)
}
```

Worked example

We simulate a 30×30 grid with 50% initial cooperators for 100 generations, using the canonical PD payoffs ($T = 5$, $R = 3$, $P = 1$, $S = 0$).

```
result <- run_spatial_pd(n = 30, generations = 100, prob_coop = 0.5, seed = 42)
```

```
snapshot_gens <- c(0, 5, 20, 100)
snapshot_df <- map_dfr(snapshot_gens, function(g) {
  grid_to_df(result$history[[g + 1]], g)
})
snapshot_df$generation <- factor(
  snapshot_df$generation,
  levels = snapshot_gens,
  labels = paste("Generation", snapshot_gens)
)

p_snapshots <- ggplot(snapshot_df,
                      aes(x = col, y = row, fill = strategy)) +
  geom_tile() +
  facet_wrap(~generation, ncol = 2) +
  scale_fill_manual(values = c("C" = okabe_ito[1], "D" = okabe_ito[5]),
                    labels = c("C" = "Cooperate", "D" = "Defect"),
                    name = "Strategy") +
  coord_equal() +
  theme_publication() +
  theme(axis.text = element_blank(),
        axis.ticks = element_blank(),
        panel.grid.major = element_blank()) +
  labs(title = "Spatial Prisoner's Dilemma: Grid Snapshots",
       x = NULL, y = NULL)

p_snapshots
```

```
save_pub_fig(p_snapshots, "spatial-pd-snapshots", width = 8, height = 8)
```

The initial random arrangement (generation 0) shows a salt-and-pepper mixture. By generation 5, cooperators have begun to aggregate into clusters while isolated cooperators surrounded by defectors have been converted. By generation 20, the pattern has largely stabilized: compact cooperator clusters persist, separated by defector boundaries. The clusters are not static — their edges shift over time — but the overall cooperation level remains roughly stable.

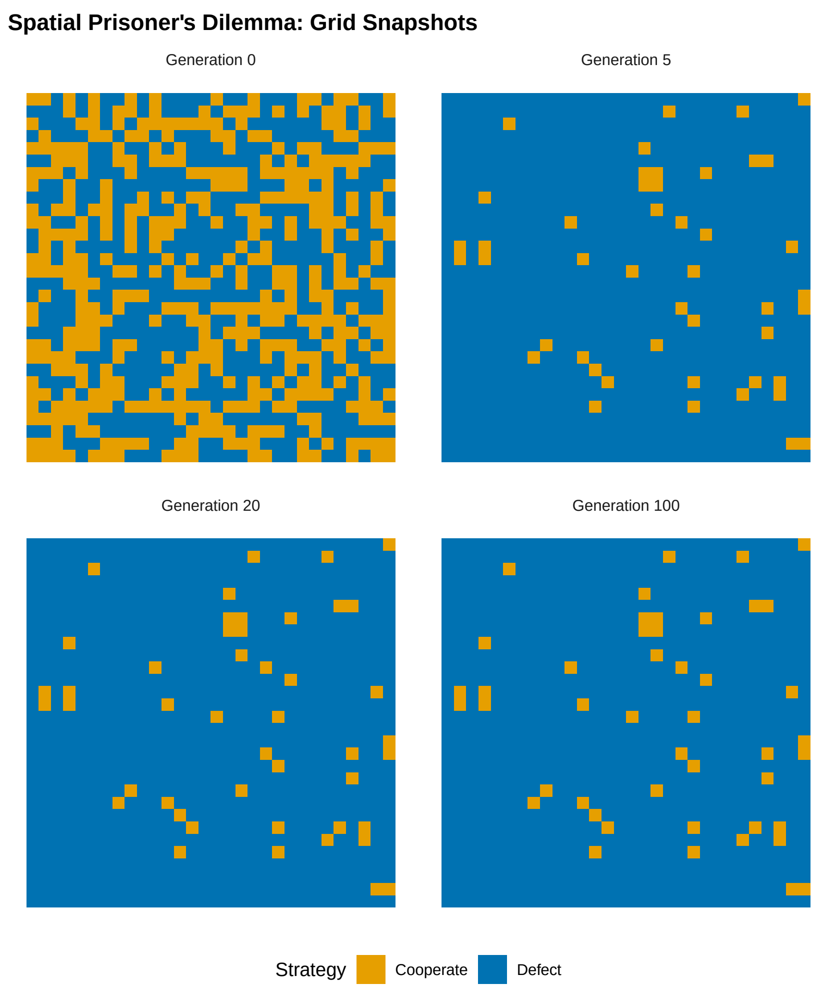

Figure 1: Grid snapshots at generations 0, 5, 20, and 100. Cooperators (orange) form compact clusters that resist invasion by defectors (blue). By generation 20, the spatial pattern has largely stabilized into a dynamic equilibrium.

```r
coop_df <- tibble(
  generation = 0:100,
  cooperation = result$coop_frac
)

p_coop <- ggplot(coop_df, aes(x = generation, y = cooperation)) +
  geom_line(colour = okabe_ito[1], linewidth = 1) +
  geom_hline(yintercept = mean(result$coop_frac[51:101]),
             linetype = "dashed", colour = okabe_ito[8]) +
  scale_y_continuous(labels = label_percent(), limits = c(0, 1)) +
  theme_publication() +
  labs(title = "Cooperation Fraction Over Time",
       x = "Generation", y = "Fraction cooperating")

p_coop
```

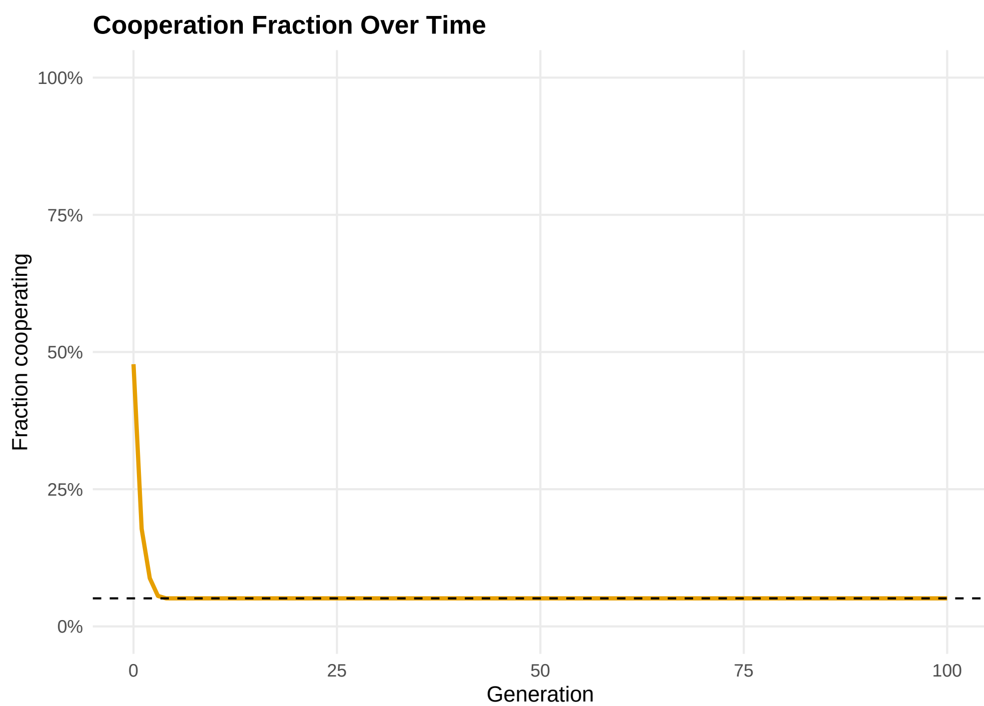

Figure 2: Cooperation fraction over 100 generations. After an initial transient where isolated cooperators are eliminated, the cooperation rate stabilizes around a dynamic equilibrium. Spatial structure sustains a substantial level of cooperation even in the one-shot Prisoner's Dilemma.

```r
save_pub_fig(p_coop, "spatial-pd-cooperation", width = 7, height = 5)
```

```r
cat(sprintf("Initial cooperation rate: %.1f%%\n",
            result$coop_frac[1] * 100))
```

```
#> Initial cooperation rate: 47.8%
```

```r
cat(sprintf("Final cooperation rate:   %.1f%%\n",
            result$coop_frac[101] * 100))
```

```
#> Final cooperation rate:   5.1%
```

```r
cat(sprintf("Mean cooperation (gen 50-100): %.1f%%\n",
            mean(result$coop_frac[51:101]) * 100))
```

```
#> Mean cooperation (gen 50-100): 5.1%
```

The cooperation rate drops sharply in the first few generations as isolated cooperators are eliminated. It then stabilizes as surviving cooperators form self-reinforcing clusters. The long-run cooperation rate depends on the payoff parameters — with the canonical values used here, spatial structure sustains cooperation at a level far above zero, even though defection would dominate in a well-mixed population.

Extensions

- **Varying temptation**: As the temptation payoff T increases (holding other payoffs fixed), the sustainable cooperation rate decreases. There is a critical threshold above which cooperators go extinct even with spatial structure. Experimenting with this threshold is a good exercise.
- **Stochastic update rules**: Instead of deterministic imitation of the best neighbor, one can use probabilistic rules where the probability of adopting a neighbor's strategy increases with the neighbor's payoff advantage. This connects to the **Fermi update rule** and reduces sensitivity to small payoff differences.
- **Network structure**: The lattice is a special case of a network. Cooperation dynamics on random graphs, scale-free networks, and small-world networks exhibit qualitatively different behavior (@ref(sec-network-games)).
- **Evolutionary stability**: The spatial model shows that cooperation can be an evolutionarily stable configuration even when it is not an ESS in the well-mixed case (@ref(sec-ess)).
- For the original model, see @nowak1992. For a comprehensive review of spatial evolutionary games, see @szabo2007.

Exercises

1. **Effect of grid size.** Run the spatial PD on grids of size 10×10, 30×30, and 50×50 with the same initial cooperation probability (50%). How does the long-run cooperation rate depend on grid size? Explain why boundary effects matter more on smaller grids.

2. **Temptation threshold.** Fix $R = 3$, $P = 1$, $S = 0$ and vary T from 3.5 to 8 in increments of 0.5. For each value, run the simulation for 100 generations and record the final cooperation rate. Plot cooperation rate vs. T and identify the approximate threshold where cooperators go extinct. (*Hint*: modify `play_round()` to accept custom payoffs.)

3. **Initial cluster.** Instead of random initial placement, start with a single 5×5 block of cooperators in the center of a 30×30 grid of defectors. Does the cooperator block survive? Grow? How does the outcome depend on T?

Solutions appear in @ref(sec-solutions).